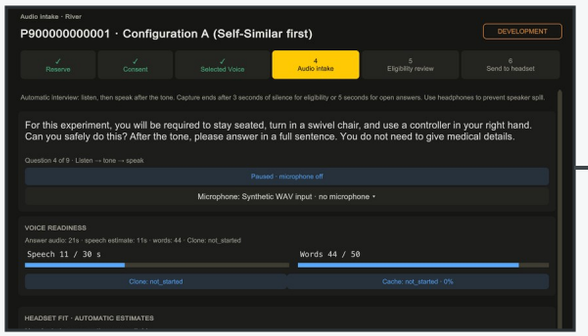
Before the session



Audio Intake
(Nine spoken prompts, Trial Review)

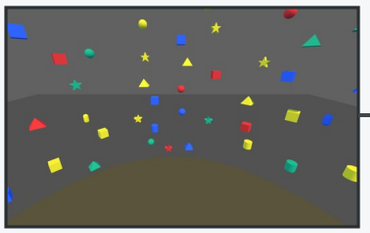
Clone and Synthesize
(ElevenLabs)



Selected Agent Voice **Self-Similar Voice**

Prepared Phrase Libraries
(Sent to the headset ahead of time)

During each trial



Search Scene
(Unity on the HTC VIVE Focus Vision)

Gaze, Head, Controller
(Headset sensors)

Guidance State

- Gaze valid and recent
- Angle to target, trend
- Direction to target
- Recent gaze direction
- Last cue and response
- Past cues and trials
- Allowed strategies

Summarize, Then Ask
(Unity on the headset)

Compact state, not raw gaze

Wireless ADB link

Two Decisions

1. Strategy
Choose one allowed: stay silent, orient, refine, correct, or encourage

2. Phrase
Rank the prepared phrases using their recent use

Decision Model
(Local MiniLM on the lab computer)

On our computer, no cloud

Chosen phrase

Spoken Cue
(Assigned voice)

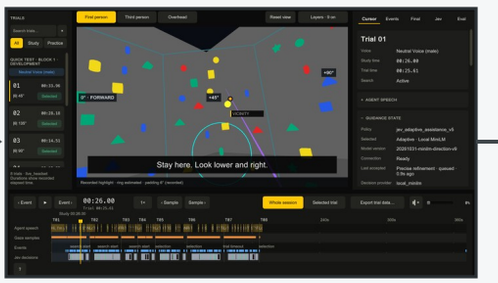
Logged every frame
Starting angle θ_0
saved each trial

What actually played feeds back into memory

After the session

Session Recordings
(Gaze, events, replay)

Transfer



Trial Review
(Replay and live monitor)

Export

Trial CSVs